

Medical AI Models - Complete Presentation Guide

Skin Cancer

Architecture: Custom CNN

Framework: TensorFlow

Dataset: ISIC (25000 images)

Input: 28x28 RGB Dermoscopy

Accuracy: 96.2%

Classes: AK,BCC,BKL,DF,MEL,NV,SCC,VASC

Training Challenges: Class imbalance, overfitting, lighting variation

Why this model: Fast, lightweight, image feature extraction

Output: Classification + confidence

Breast Cancer

Architecture: U-Net + MobileNetV2

Framework: TensorFlow

Dataset: BUSI (15000 images)

Input: Ultrasound

Accuracy: 96.7%

Classes: Benign, Malignant, Normal

Training Challenges: Mask annotation, speckle noise

Why this model: Segmentation then classification

Output: Mask + class

Eye Disease

Architecture: EfficientNetB3

Framework: TensorFlow

Dataset: Fundus (15000 images)

Input: 224x224 Fundus

Accuracy: 97.8%

Classes: Cataract, DR, Glaucoma, Normal

Training Challenges: Image quality, reflections

Why this model: Transfer learning

Output: Disease report

Brain Tumor

Architecture: VGG-like CNN

Framework: TensorFlow

Dataset: MRI (7000 images)

Input: 224x224 MRI

Accuracy: 98.1%

Classes: Glioma, Meningioma, Pituitary, No Tumor

Training Challenges: Overfitting, memory

Why this model: Deep features

Output: Tumor type

Heart

Architecture: U-Net

Framework: TensorFlow

Dataset: Heart (13000 images)

Input: 128x128 Echo

Accuracy: 97.3%

Classes: Normal, Slightly Large, Abnormal

Training Challenges: Low contrast

Why this model: Best for segmentation

Output: Mask+ratio

Chest X-Ray

Architecture: ResNet152+EfficientNetB5+Attention

Framework: PyTorch

Dataset: Chest X-Ray (22000 images)

Input: 224x224 X-Ray

Accuracy: 96.9%

Classes: COVID, Viral, Bacterial, TB, Emphysema, Normal

Training Challenges: Large model, class imbalance

Why this model: Hybrid accuracy

Output: Disease

Kidney

Architecture: Custom CNN

Framework: TensorFlow

Dataset: Kidney CT (8000 images)

Input: 200x200 CT

Accuracy: 97.5%

Classes: Cyst, Stone, Tumor, Normal

Training Challenges: Similar classes, CT quality

Why this model: Fast and accurate

Output: Disease